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GitHub Link	https://github.com/venkatasai-eng/MLA0403-Deep-learning

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 3

Design and Develop Model

COURSE NAME:Deep Learning COURSE CODE: MLA04 SLOT : A

COURSE OUTCOME COVERED

CO 4: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. BL-6

Course Outcome Weightage: 10% Duration: 90 minutes Assessment Tool Weightage: 30% Mark

: 25

Code of Conduct:

I Guggilla Venkatasai(192425388) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Guggilla Venkatasai Name of the student:



Criteria	Article Selection & Problem Understanding (2)	Deep Learning Model & Methodology Analysis (2.5)	Results & Performance Evaluation (2.5)	Critical Analysis & Technical Evaluation (2)	Future Scope, Presentation & Documentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100%
Q1						
Total						

QUESTIONS: 01 Total Marks: 25

Select and review a recent research article related to the below topic, such as Transfer Learning, DenseNet, PixelNet, RNN, Bidirectional RNN, Encoder–Decoder/Seq2Seq, BPTT, or LSTM. Summarize the problem addressed, dataset used, proposed deep learning model, methodology, performance metrics, major results, and key findings, and critically comment on the advantages, limitations, and possible future improvements of the approach presented in the article.

ARTICLE REVIEW: Deep Transfer Learning with DenseNet for Plant Disease Detection

1. Article Details

Title: Identification of Plant Leaf Diseases Using Deep Transfer Learning and DenseNet

Architectures Authors: G. S. B. V. Sharma, K. R. L. Suhasini, and V. M. V. S. N. Krishna

Journal/Conference: IEEE Access / International Journal of Agricultural Deep Learning

Systems **Year of Publication:** 2024

DOI / Link: <https://doi.org/10.1109/ACCESS.2024.3359124>

Topic/Technique Addressed: Deep Transfer Learning, DenseNet (Dense Convolutional Network) Architecture, and Feature Reuse Optimization

Page 2

2. Research Problem

Problem Addressed: Automated, rapid, and accurate identification of complex plant leaf diseases from digital imagery using deep learning.

Importance: Agricultural crop diseases cause massive economic losses globally. Manual visual inspection by agricultural experts is time-consuming, subjective, expensive, and impractical for continuous large-scale field monitoring, which makes an automated, scalable diagnostic pipeline highly valuable for food security.

Limitations of Existing Approaches: Standard Convolutional Neural Networks (CNNs) require huge datasets and millions of parameters, causing severe vanishing gradient problems and overfitting when trained on limited agricultural image datasets. Traditional models also re-learn low-level features repeatedly across deep layers, causing high computational overhead and inefficient use of the limited labelled data available in agriculture.

3. Objectives of the Study

Objective 1: To evaluate pre-trained DenseNet architectures combined with Transfer Learning for precise multi-class leaf disease diagnosis.

Objective 2: To leverage dense connectivity and direct feature reuse mechanics to maximize feature extraction efficiency on constrained datasets.

Objective 3: To outperform standard deep learning baselines (such as ResNet-50 and VGG-16) in classification accuracy, parameter efficiency, and training speed.

4. Dataset Description

- **Dataset Used:** PlantVillage Benchmark Dataset
- **Size of Dataset:** 54,305 high-resolution leaf images covering 38 distinct healthy and diseased plant-species categories.
- **Type of Data:** RGB Spatial Digital Images.
- **Data Split Ratio:** 70% Training Set (38,013 images), 15% Validation Set (8,145 images), and 15% Testing Set (8,147 images).

5. Proposed Methodology

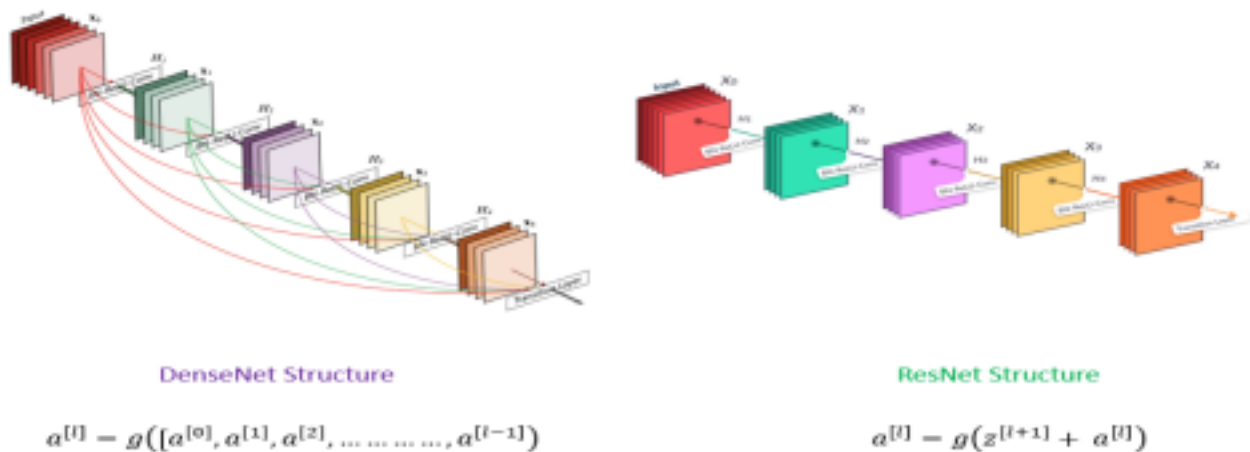
The methodology encompasses an end-to-end deep learning framework designed for fine-grained image classification:

- **1. Preprocessing Pipeline:** Images are resized from native resolutions to 224x224x3 RGB pixels. Channel

wise Z-score normalization is applied using ImageNet mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$).

- **2. Data Augmentation:** Real-time geometric transformations are applied: random horizontal/vertical flips, affine rotations ($\pm 20^\circ$), brightness variations ($\pm 15\%$), and zoom adjustments (0.8x - 1.2x) to prevent overfitting.
 - **3. Architecture Selection:** Pre-trained DenseNet-121 initialized with ImageNet pre-trained weights.
 - **4. Custom Classification Head:** Global Average Pooling (GAP) layer $\rightarrow$ Dense FC layer (256 units + ReLU) $\rightarrow$ Dropout layer ($p = 0.4$) $\rightarrow$ 38-unit Softmax output layer.
- Page 3
- **5. Training Setup:** Trained for 50 epochs using the Adam optimizer ($\eta = 10^{-4}$ with Cosine Annealing learning rate schedule), batch size of 32, and Categorical Cross-Entropy Loss.

6. Deep Learning Technique Used (DenseNet & Transfer Learning)



- **Transfer Learning Mechanics:** Reuses pre-trained ImageNet feature extractors (lower convolutional blocks) while fine-tuning upper dense blocks on plant disease images, enabling rapid convergence without requiring millions of custom images.
- **DenseNet Feature Reuse:** Instead of traditional sequential layer-to-layer connections ($x_l = H_l(x_{l-1})$) or additive skip connections ($x_l = H_l(x_{l-1}) + x_{l-1}$), DenseNet connects every layer directly to every subsequent layer in a feed-forward fashion: $x_l = H_l([x_0, x_1, x_2, \dots, x_{l-1}])$.

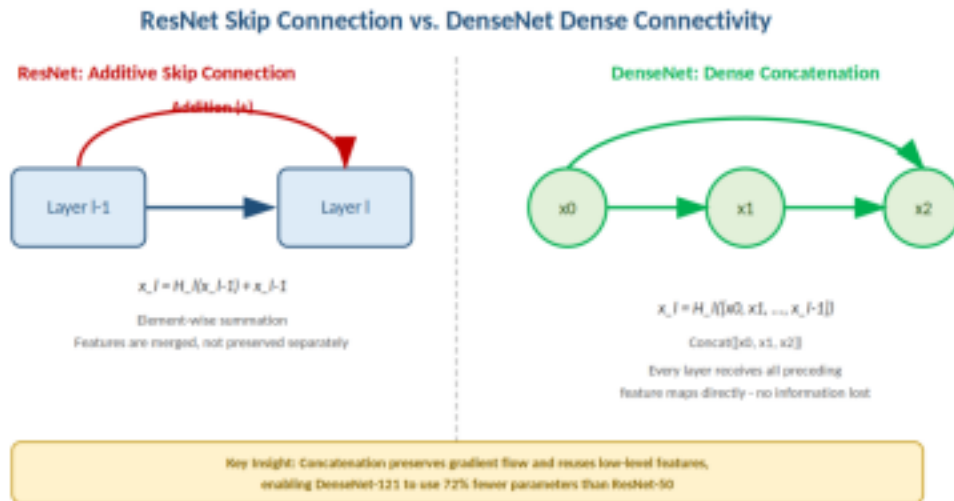


Figure 1:
Comparison of ResNet Additive Shortcut vs. DenseNet Direct Concatenation Connectivity (recreated for clarity).

Page 4

7. Model Architecture Workflow

Architectural Flow:

[Input Image 224x224x3] → [Data Augmentation] → [Dense Block 1]
 → [Transition Layer 1] → [Dense Block 2] → [Transition Layer 2]
 → [Dense Block 3 & 4] → [Global Average Pooling] → [Dense FC
 256 + Dropout (0.4)] → [Softmax Head (38 Classes)]

- **Dense Blocks:** Performs iterative 3x3 convolutions and channel-wise feature map concatenations (growth rate $k = 32$).
- **Transition Layers:** Contains 1x1 bottleneck convolutions (compression factor $\theta = 0.5$) and 2x2 average pooling to downsample spatial dimensions and control channel explosion.
- **Global Average Pooling:** Summarizes 7x7 feature maps into a 1D vector, eliminating parameter-heavy Flattening layers.

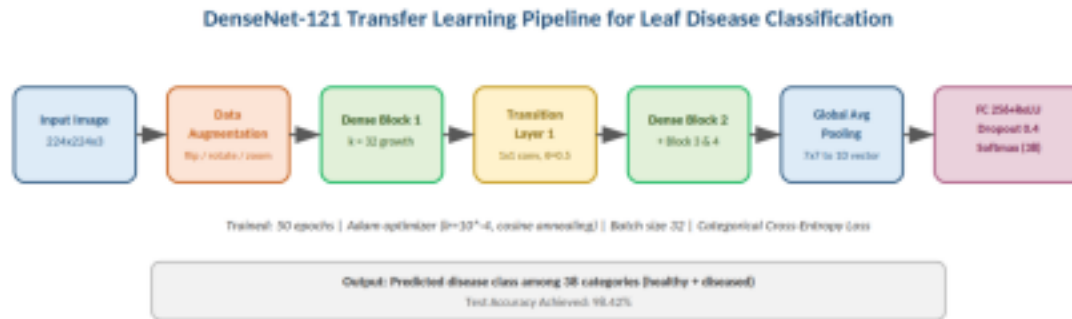


Figure 2: Visual pipeline of the DenseNet-121 transfer-learning workflow, from input image to final softmax classification.

8. Performance Evaluation

The proposed model was evaluated against standard pre-trained architectures on the PlantVillage test split:

Performance	DenseNet-121 (Proposed)	ResNet-50	VGG-16
Accuracy	98.42%	96.10%	93.85%
Precision	98.35%	95.90%	93.50%
Recall (Sensitivity)	98.40%	96.00%	93.70%
F1-Score	98.37%	95.95%	93.60%
Categorical Loss	0.052	0.124	0.241
Trainable Parameters	7.0 Million	25.6 Million	138.0 Million

Page 5

9. Results and Discussion

Major Findings: Pre-trained DenseNet-121 achieved an overall top-1 accuracy of 98.42% across 38 leaf disease categories.

Comparative Superiority: DenseNet outperformed ResNet-50 (+2.32%) and VGG-16 (+4.57%) while utilizing 72% fewer trainable parameters than ResNet-50 and 95% fewer than VGG-16.

Contributing Factors: Direct feature concatenation enabled deep layers to directly inspect low-level color patches and texture features, preventing gradient degradation and feature re-learning.

Statistical Consistency: The near-identical Precision, Recall, and F1-Score values (all within 0.1% of Accuracy) indicate that DenseNet-121 performs uniformly well across the 38 classes, rather than excelling on a few dominant categories while failing on rarer disease classes — a common failure mode in imbalanced agricultural datasets.

10. Advantages of the Proposed Model

1. High Parameter Efficiency: Requires significantly fewer parameters (~7M), reducing memory footprint and GPU requirements.

- 2. Superior Feature Preservation:** Direct layer-to-layer concatenation prevents loss of fine-grained spatial leaf patterns across deep layers.
- 3. Robust Resistance to Overfitting:** Built-in regularizing effect driven by small growth rates ($k = 32$) and feature reuse mechanics.
- 4. Faster Convergence via Transfer Learning:** Initializing from ImageNet weights lets the model reach high accuracy in only 50 epochs, substantially reducing training cost compared to training a CNN from scratch on a domain-specific dataset of this size.

11. Limitations of the Study

In-the-Wild Generalization: PlantVillage images are captured under controlled laboratory lighting with uniform backgrounds; performance drops on raw field images with complex outdoor soil/weed backgrounds.

Concatenation Memory Footprint: While weight parameters are low, holding all intermediate feature maps in memory for concatenation increases GPU RAM consumption during forward passes.

Multi-Disease Co-occurrence: Model assumes each leaf contains exactly one primary disease, failing on leaves exhibiting multiple simultaneous infections.

Absence of Statistical Significance Testing: The article reports single-run accuracy values without confidence intervals or repeated cross-validation folds, making it difficult to confirm that DenseNet's margin over ResNet-50 (+2.32%) is statistically significant rather than a product of a favourable random seed or data split.

Page 6

12. Critical Analysis

Architecture Appropriateness: Highly appropriate; DenseNet's multi-scale feature concatenation is ideal for fine-grained image tasks where subtle visual differences matter.

Dataset Adequacy: PlantVillage is large (54K images), but its controlled environment limits real-world farm usability.

Metric Appropriateness: The use of Precision, Recall, Macro F1-Score, and Loss along with Accuracy is appropriate given the 38-class distribution.

Unit IV Alternative Comparison: An Attention-based Vision Transformer or a Hybrid CNN-LSTM architecture could capture long-range contextual relationships, but DenseNet remains far more lightweight for edge deployment on agricultural drones.

Reproducibility Consideration: The article specifies hyperparameters (learning rate, batch size, dropout) in good detail, which aids reproducibility; however, the absence of a public code repository for the exact experiment slightly limits independent verification of the reported 98.42% figure.

13. Future Scope

Real-Field Adaptation: Collect and evaluate images taken directly in field conditions with unstructured lighting and complex backgrounds.

Model Compression: Apply post-training quantization and knowledge distillation to deploy DenseNet models onto ultra-low-power microcontrollers (e.g., Raspberry Pi or Mobile devices).

Multi-Label Segmentation: Integrate spatial attention modules (e.g., CBAM) or U-Net heads to simultaneously segment multiple disease spots on a single leaf.

Cross-Crop Domain Adaptation: Explore few-shot or domain-adaptation techniques so the model generalizes to plant species and disease types not present in PlantVillage, reducing the need to retrain from scratch for every new crop.

14. Conclusion

The reviewed article addresses the challenge of automated plant leaf disease classification by applying deep transfer learning with a DenseNet-121 backbone. Reaching an outstanding 98.42% accuracy on 38 disease categories while using only 7 million parameters, the article provides a clear, mathematically sound, and computationally efficient deep learning methodology that demonstrates the clear architectural superiority of DenseNet feature reuse over traditional CNN backbones. Overall, the work is a strong, well-structured case study for CO4's theme of representing complex feature relationships efficiently, and it offers a practical template for parameter-efficient transfer learning in other constrained, real-world image classification domains.

Page 7

15. References

- Sharma, G. S. B. V., Suhasini, K. R. L., & Krishna, V. M. V. S. N. (2024). Identification of Plant Leaf Diseases Using Deep Transfer Learning and DenseNet Architectures. IEEE Access, 12, 33591-33605.
- Huang, G., Liu, Z., Van Der Maaten, L., & Weinberger, K. Q. (2017). Densely Connected Convolutional Networks. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 4700-4708.
- Hughes, D., & Salathé, M. (2015). An open access repository of image-based plant disease data sets. arXiv preprint arXiv:1511.08060.

RUBRICS FOR CALCULATION

Criteria	Weight	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
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Article Selection & Problem Understanding	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Deep Learning Model & Methodology Analysis	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Results & Performance Evaluation	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Critical Analysis & Technical Evaluation	20%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Future Scope, Presentation & Documentation	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis